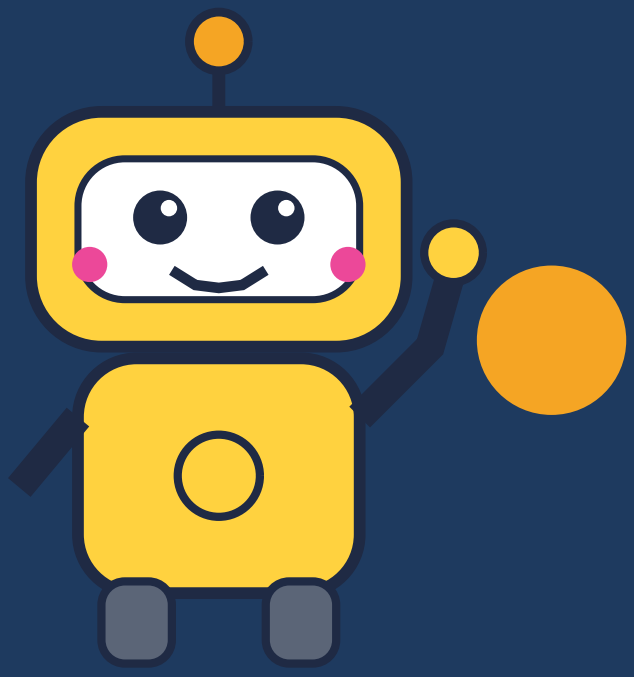
SAMPLE PAPERS

Think Like a Coder

Sample examination papers, Class 4

Half-yearly and annual papers
40 marks, 1½ hours, with blueprints and marking schemes



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 4 · Photocopiable for school use

SAMPLE EDITION · DRAFT

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About these papers

These are two sample examination papers for Class 4: a **half-yearly** paper on the term-1 chapters and an **annual** paper on the whole year. They go with *Think Like a Coder*, Class 4, and with the chapter assessments. All questions are new.

How they follow CBSE

- CBSE does not hold board examinations for Classes 3-8 and has not published a question-paper pattern for CT&AI.; Schools set their own papers. These papers follow what CBSE's CT&AI; handbooks describe: **competency-based** written tests with CT puzzles, with no 'write a program' questions and no computer needed.
- The question types are the ones CBSE uses in its competency-based papers: multiple choice, short answers and case-based questions.
- Questions are spread across four thinking levels (remember, understand, apply, analyse and create), with most marks for applying and analysing. The blueprints show the split.
- Chapters follow the order of CBSE's 2026-27 CT&AI; handbooks and the NCERT maths chapters.

Paper design

Section	Question type	Questions	Marks
A	Multiple choice	8	8 x 1 = 8
B	Very short answer	6	6 x 2 = 12
C	Short answer	4	4 x 3 = 12
D	Case-based questions	2	2 x 4 = 8
	Total · 1½ hours	20	40

Using the papers

- Use the half-yearly paper in September and the annual paper in February-March, or as mock tests before the school's own exams.
- Print one paper per student; the papers have space for answers.
- Read questions aloud to students who find reading hard: the paper tests thinking, not reading.
- Mark with the marking scheme; accept any correct method, and give part marks for correct steps.
- Use the blueprints to set your own papers in the same pattern.

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SAMPLE PAPER · HALF-YEARLY EXAMINATION

Class 4 · Computational Thinking

Syllabus: chapters 1-7 (term 1)

Time allowed: 1½ hours **Maximum marks:** 40

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 20 questions in 4 sections. All questions are compulsory.
   Section A: multiple choice, 8 questions of 1 mark each; Section B: very short answer, 6 questions of 2 marks each;
2. Section C: short answer, 4 questions of 3 marks each; Section D: case-based questions, 2 questions of 4 marks each.
3. Marks for each question are shown on the right.
4. Write your answers in the space given below each question.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice — 8 x 1 = 8 marks

1. How many edges does a cube have? [1]

a) 6 **b)** 8 **c)** 12

2. Rule: if it is sunny then wear a hat, else carry an umbrella. What do you do on a rainy day? [1]

a) wear a hat **b)** carry an umbrella

3. Which number is even? [1]

a) 35 **b)** 47 **c)** 58

4. Even + odd is always ... [1]

a) even **b)** odd

5. 6,000 + 400 + 9 = ... [1]

a) 6,409 **b)** 6,490 **c)** 64,009

6. One quarter of 16 is ... [1]

a) 2 **b)** 4 **c)** 8

7. 250 cm = ... [1]

a) 2 m 5 cm **b)** 2 m 50 cm **c)** 25 m

8. A fan can be ON or OFF. This is the fan's ... [1]

a) state **b)** shape **c)** size

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SECTION B · Very short answer — 6 x 2 = 12 marks

9. How many faces, edges and corners does a matchbox (a cuboid) have? [2]

10. Rule: if the number is even then add 10, else add 1. What comes out for 6, and for 9? [2]

11. Using 3, 0, 7 and 5 once each, write the biggest and the smallest 4-digit numbers. [2]

12. A roti is cut into 4 pieces of different sizes. Is each piece a quarter? Why? [2]

13. Robo faces right. It turns left, then turns left again. Which way does it face now? [2]

14. A toggle light starts ON. You press it 5 times. Is it ON or OFF? Why? [2]

SECTION C · Short answer — 4 x 3 = 12 marks

15. Add one check coin to the end of each row so that every row has an even number of black coins. Row 1: black, white, black. Row 2: white, white, black. Row 3: black, black, black. Write the colour of each check coin. [3]

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16. Sort 8, 3 and 5 with the sorting network (top two wires, then bottom two, then top two). Write the order after each comparator. [3]

17. Write a loop with turns that draws a square with sides of 5 steps. How many steps does Robo walk in all? [3]

18. Write exact steps to draw a square with sides of 3 cm and both of its diagonals. [3]

SECTION D · Case-based questions — 2 x 4 = 8 marks

19. A robot sorts things on a belt: if the thing is round then say 'R' and put it in basket R, else say 'L' and put it in basket L. The things are a ball, a pencil, a coin and a ruler.

(a) Where does the ball go? [1]

(b) Which things go into basket L? [1]

(c) The robot runs: repeat 4 times: pick the next thing, then use the rule. The things come in this order: coin, ruler, ball, pencil. What does it say? [2]

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20. A school gate has a light. Green means open, red means closed. Button G makes it green, button R makes it red, and button T switches it to the other colour. The light starts red.

(a) Press T. What colour is the light? [1]

(b) Start again from red. Press T, T, G. What colour is it now? [1]

(c) Start again from red. Make a trace table for R, T, T, T. Is the gate open or closed at the end? [2]

***** End of the question paper *****

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SAMPLE PAPER · ANNUAL EXAMINATION

Class 4 · Computational Thinking

Syllabus: all chapters, mostly chapters 8-14 (term 2)

Time allowed: 1½ hours **Maximum marks:** 40

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 20 questions in 4 sections. All questions are compulsory.
   Section A: multiple choice, 8 questions of 1 mark each; Section B: very short answer, 6 questions of 2 marks each;
2. Section C: short answer, 4 questions of 3 marks each; Section D: case-based questions, 2 questions of 4 marks each.
3. Marks for each question are shown on the right.
4. Write your answers in the space given below each question.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice — 8 x 1 = 8 marks

1. Which unit measures how heavy a bag is? [1]

a) kilogram **b)** litre **c)** metre

2. Outer loop: repeat 3 times. Inner loop: repeat 6 times: clap. How many claps? [1]

a) 9 **b)** 18 **c)** 36

3. In a clue table, a tick means ... [1]

a) it is true **b)** it is not possible

c) not sure

4. Which shape has exactly one line of symmetry? [1]

a) a heart shape **b)** a circle **c)** a square

5. Start at 5:00. Repeat until the clock shows 7:00: add 30 minutes. How many times does the loop run? [1]

a) 2 **b)** 4 **c)** 6

6. In a decision tree, each answer takes you along a ... [1]

a) branch **b)** loop **c)** clock

7. Two groups of five tally marks and three more marks show ... [1]

a) 8 **b)** 13 **c)** 23

8. 1 m 5 cm = ... cm [1]

a) 15 **b)** 105 **c)** 150

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SECTION B · Very short answer — 6 x 2 = 12 marks

9. Fill in: 2 kg 300 g = ___ g, and 3 L 50 mL = ___ mL. [2]

10. Write a nested loop to plant 4 rows of 5 trees. How many trees are planted? [2]

11. Write three capital letters that have a line of symmetry going straight down. [2]

12. Today is Saturday. What day was it 9 days ago? [2]

13. Write one yes/no question that separates a cow from a hen. [2]

14. Is 47 + 38 odd or even? Explain without adding. [2]

SECTION C · Short answer — 4 x 3 = 12 marks

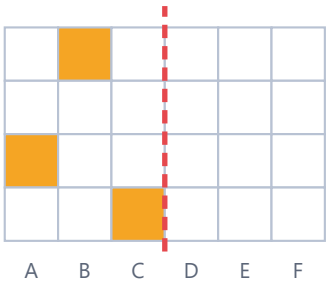
15. Tara, Uday and Vani sit in seats 1, 2 and 3. Tara is not in seat 1. Uday is in seat 3. Who sits where? [3]

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16. You have a 5 litre jug and a 2 litre jug. Write the steps to get exactly 1 litre. [3]

17. Write a repeat-until loop for saving ₹10 every week until you have ₹60. How many weeks does it take? [3]

18. The red dashed line is a mirror. Shaded squares are in columns A, B and C. In which column is the mirror partner of each one? [3]



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SECTION D · Case-based questions **2 x 4 = 8 marks**

19. The bar graph shows how the children of Class 4 come to school.



(a) Which way is the most common? [1]

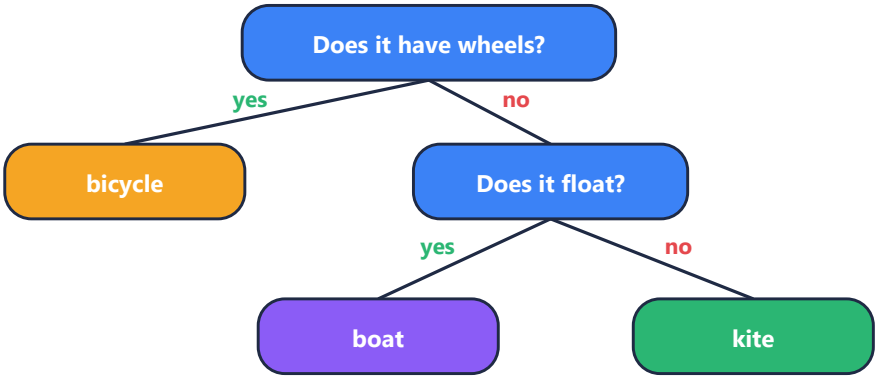
(b) How many more children walk than come by car? [1]

(c) How many children are there in all? [1]

(d) Draw the tally marks for the children who cycle. [1]

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20. Look at this decision tree.



(a) Where does a car end up? [1]

(b) Why is that wrong? What question would you add to fix the tree? [2]

(c) Where does a kite end up? [1]

***** End of the question paper *****

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Blueprints

Blueprint: Half-yearly examination

Syllabus: chapters 1-7 (term 1). Marks by chapter and section.

Chapter	A	B	C	D	Marks
1. Shapes Around Us	1	2	3		6
2. Hide and Seek	1	2		4	7
3. Patterns Around Us	2		3		5
4. Thousands Around Us	1	2	3		6
5. Sharing and Measuring	1	2			3
6. Measuring Length	1	2	3		6
7. The Cleanest Village	1	2		4	7
Total	8	12	12	8	40

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	2	5%
Understand	explain or interpret an idea	6	15%
Apply	use an idea or rule in a new problem	21	52%
Analyse and create	trace, debug, compare, design or justify	11	28%

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Blueprint: Annual examination

Syllabus: all chapters, mostly chapters 8-14 (term 2). Marks by chapter and section.

Chapter	A	B	C	D	Marks
3. Patterns Around Us		2			2
6. Measuring Length	1				1
8. Weigh It, Pour It	1	2	3		6
9. Equal Groups	1	2			3
10. Elephants, Tigers and Leopards	1		3		4
11. Fun with Symmetry	1	2	3		6
12. Ticking Clocks and Turning Calendar	1	2	3		6
13. The Transport Museum	1	2		4	7
14. Data Handling	1			4	5
Total	8	12	12	8	40

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	3	8%
Understand	explain or interpret an idea	1	2%
Apply	use an idea or rule in a new problem	24	60%
Analyse and create	trace, debug, compare, design or justify	12	30%

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Marking schemes

Accept any correct method and any sensible wording. Give part marks for correct steps, even when the final answer has a slip.

Marking scheme: Half-yearly examination

1. c) 12
2. b) carry an umbrella
3. c) 58
4. b) odd
5. a) 6,409
6. b) 4
7. b) 2 m 50 cm
8. a) state
9. 6 faces, 12 edges, 8 corners (2 marks for all three; 1 mark for two)
10. 16 and 10 (1 mark for each)
11. 7,530 and 3,057 (1 mark for each (0 cannot come first))
12. No, because quarters must be 4 equal parts (1 mark for no, 1 mark for the reason)
13. left (2 marks; 1 mark for 'up' after the first turn)
14. OFF, because an odd number of presses flips it (1 mark for OFF, 1 mark for the reason)
15. Row 1: white. Row 2: black. Row 3: black. (1 mark for each row)
16. After 1: 3, 8, 5. After 2: 3, 5, 8. After 3: 3, 5, 8. (1 mark for each comparator)
17. Repeat 4 times: forward 5, turn right. 20 steps. (2 marks for the loop, 1 mark for 20)
18. 1. Draw a line AB 3 cm long. 2. At A and at B draw lines 3 cm long, straight up. 3. Join the tops to finish the square. 4. Join each corner to the opposite corner. (1 mark for exact lengths, 1 mark for the square, 1 mark for the diagonals)
19. (a) basket R (b) the pencil and the ruler (c) R, L, R, L ((a) 1 mark (b) 1 mark (c) 2 marks, 1 mark with one slip)
20. (a) green (b) green (c) R: red; T: green; T: red; T: green. The gate ends open. ((a) 1 mark (b) 1 mark (c) 2 marks for a full trace, 1 mark for two correct rows)

Marking scheme: Annual examination

1. a) kilogram
2. b) 18
3. a) it is true
4. a) a heart shape
5. b) 4
6. a) branch
7. b) 13
8. b) 105
9. 2,300 g; 3,050 mL (1 mark for each)
10. Repeat 4 times: (repeat 5 times: plant a tree). 20 trees. (1 mark for the nested loop, 1 mark for 20)
11. For example A, H, M (also I, O, T, U, V, W, X, Y) (2 marks for three correct letters; 1 mark for two)
12. Thursday (2 marks; 1 mark for seeing that 7 days ago was Saturday)
13. For example: Does it have four legs? (1 mark for a yes/no question, 1 mark if it separates them)
14. Odd, because odd + even is always odd (1 mark for odd, 1 mark for the reason)
15. Vani: seat 1. Tara: seat 2. Uday: seat 3. (1 mark for each person)
16. Fill the 5 L jug. Pour from it to fill the 2 L jug (3 L left). Empty the 2 L jug. Pour again from the 5 L jug to fill the 2 L jug. 1 L is left in the 5 L jug. (3 marks for steps that work; 2 marks if one step is missing; 1 mark for a sensible start)
17. Set savings to 0. Repeat until savings = 60: add 10. It takes 6 weeks. (1 mark for the loop, 1 mark for the condition, 1 mark for 6)
18. A goes to F, B goes to E, C goes to D (each stays in its own row) (1 mark for each)
19. (a) walking (b) 2 more (c) 28 (d) four tally marks: |||| (1 mark for each part)
20. (a) bicycle (b) The tree has only one leaf for things with wheels, so a car is called a bicycle. Add the question 'Does it have an engine?' after 'yes': yes, car; no, bicycle. (c) kite ((a) 1 mark (b) 2 marks: 1 mark for the reason, 1 mark for a question that fixes it (c) 1 mark)

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